

MONDIP SEN

Senior Backup & Data Protection Engineer | Cloud, Kubernetes & Storage | IaC, Observability & AI

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SUMMARY

Senior data protection and infrastructure engineer with **20+ years** designing, **migrating**, and operating enterprise backup, storage, and disaster recovery (DR) environments for Fortune 500 clients across healthcare, financial services, energy, consumer goods, and government. STS-certified on Veritas NetBackup, with hands-on depth across the full data-protection ecosystem — **NetBackup, Commvault, Veeam, Rubrik, Cohesity, IBM Spectrum Protect (TSM), Dell EMC NetWorker, Avamar, and PowerProtect DD** — protecting Linux, UNIX, Windows, VMware, and **AWS / Azure / GCP / OCI** workloads. Former **global data-protection team lead**, now a **Solutions Architect** delivering cloud-native, automated, observability-driven infrastructure with **Kubernetes, Terraform, Ansible, and CI/CD**.

CORE TECHNICAL SKILLS

- **Enterprise Backup & Recovery:** Veritas NetBackup 6.x–10.x (STS-certified), NetBackup Appliances (5230/5330), IBM Spectrum Protect (TSM), Dell EMC NetWorker, Avamar, Data Domain / **PowerProtect DD**, Veeam Backup & Replication, Commvault, **Rubrik, Cohesity**, NetApp AltaVault, Veritas Backup Exec
- **Backup Engineering & Migration:** Platform **migration** and upgrade (NetBackup 6.x→10.x, physical to cloud, tape to disk-to-cloud); Storage Lifecycle Policies, Auto Image Replication (AIR), PDDO/MSDP deduplication, multiplexing/streaming, NDMP, Granular Recovery (GRT), OpsCenter/NOM/VOM, capacity planning, performance tuning
- **Disaster Recovery & Business Continuity:** DR architecture, testing and runbooks; **RTO/RPO** definition; EMC SRDF replication; multi-site failover; BC/DR audit; ransomware recovery, immutable copies and air-gapped retention
- **Cloud Platforms & Containers:** AWS, Microsoft Azure, Google Cloud (GCP), Oracle Cloud Infrastructure (OCI); managed Kubernetes — **Amazon EKS, Azure AKS, Google GKE, Red Hat OpenShift, Helm, Docker**; AWS Backup, Azure Backup / Site Recovery; EC2/S3/EBS, MinIO; cloud HA and failover architecture
- **Infrastructure as Code & Automation:** **Terraform, Ansible**, Chef, Puppet; **GitOps** — Git as source of truth; **GitLab CI, Jenkins**, GitHub Actions; Python, Bash/KSH, PowerShell, SQL; Prefect workflow orchestration; systemd service engineering
- **Observability, SRE & Security:** **Datadog, Splunk, IBM Instana**, Prometheus, Grafana, Loki, Tempo, **OpenTelemetry**, Alertmanager; SLO and alert design; **HashiCorp Vault** (AppRole, envconsul); **Zero Trust** access, TLS/PKI; **HIPAA, PCI DSS, SEC 17a-4 / WORM** retention, **NIST & CIS Benchmarks**
- **Storage, SAN/NAS & Virtualization:** Dell EMC VMAX / **PowerMax** / VNX / Unity, NetApp FAS / **ONTAP** / WAFL, Hitachi, HP, **TrueNAS**; Brocade and Cisco Nexus SAN fabrics; LUN provisioning, multipathing, NFS/CIFS/iSCSI, Fibre Channel, ZFS; VMware vSphere / ESXi / vCenter / vMotion, Microsoft Hyper-V, **Proxmox VE, OpenStack**
- **Operating Systems & Databases Protected:** Red Hat Enterprise Linux (RHEL), SUSE, Ubuntu, IBM AIX, HP-UX, Solaris, Windows Server 2008–2022; Oracle (RMAN), Microsoft SQL Server, Exchange, SharePoint, SAP, DB2, Sybase, Active Directory
- **Process & Tooling:** **Agile / Scrum, Jira, Confluence**, ITIL, ServiceNow, BMC Remedy, Zendesk; sprint planning and backlog grooming, vendor and project management, service operational procedures, technical documentation

PROFESSIONAL EXPERIENCE

Solutions Architect — Bell Integration

New York, NY (Remote) · Dec 2024 – Present

Solutions architecture across enterprise infrastructure, cloud, storage modernization, and data protection for clients including **Chemours, American Standard, and LIXIL**.

- **Architect and operate enterprise microservices and platform infrastructure** on Docker and **Kubernetes** — 16 services spanning HashiCorp Vault, Prefect, Postgres, Redis, NATS, and MinIO — secured end-to-end with **zero-trust network access**, TLS everywhere, SSO, and **Vault AppRole** secrets management across 18 secret paths.
- **Designed comprehensive observability and SRE framework:** Prometheus metrics, Grafana dashboards, Loki log aggregation, Tempo distributed tracing, **OpenTelemetry** instrumentation, and Alertmanager paging, with SLO-based alerting and automated runbooks.
- **Automated the entire estate as code** with **Terraform** and **Ansible**, delivered through **GitLab CI** and **Jenkins** pipelines; orchestrated **90+ Prefect workflow deployments** across 8 systemd services, all version-controlled in Git.
- **Operate multi-cloud workloads** across **AWS, Azure, GCP, and OCI** under a hash-chained, daily-sealed audit log meeting **WORM / SEC 17a-4** retention discipline, with **686 automated contract tests passing at 100%**.
- **Engineered AI infrastructure and LLM automation** in **Python** with multi-provider routing, local retrieval, and prompt-cache **cost engineering (FinOps)**; extended monitoring across **Datadog** and **Splunk**.

Global Data Protection Team Lead / Sr. Data Protection Engineer — IPsoft (Amelia US LLC)

New York, NY · Mar 2016 – Nov 2024

Enterprise backup and DR for Fortune 500 clients — **Estée Lauder, McKesson, Moody's, Celesio EU, Duracell, BP, and Sands Resorts** — across healthcare, financial services, energy, consumer goods, US federal, and cloud-provider sectors.

- **Led the global data-protection team** in an **Agile/Scrum** delivery model tracked in **Jira** and documented in **Confluence**, owning service operational procedures, engineer onboarding and training, vendor and project management, client escalations, and daily service metrics.

- **Architected, deployed, upgraded, and operated enterprise backup estates** — Veritas NetBackup (6.x–10.x), Veeam Enterprise, IBM Spectrum Protect, Dell EMC NetWorker, Data Domain / PowerProtect DD, EMC Unity, and NetApp AltaVault — protecting RHEL, SUSE, AIX, and Windows systems running Exchange, SharePoint, SQL Server, and Oracle across on-premises, **Azure, AWS, and GCP** environments.
- **Ran quarterly business-continuity and DR audits** for enterprise clients, including historical-recovery validation and full DR failover testing on NetBackup with Dell EMC Data Domain, consistently proving **RTO < 45 min and RPO < 15 min**.
- **Resolved complex backup failures and L3 escalations** spanning NDMP, Exchange, Oracle RMAN, Sybase, SQL Server, SharePoint, and DB2; performance-tuned storage lifecycle policies and deduplication to achieve **28% reduction in storage overhead**.
- **Administered multi-vendor enterprise storage arrays** (Dell EMC PowerMax/Unity, NetApp ONTAP, Pure Storage, HP, IBM, Hitachi): LUN provisioning and decommission, multipath support, and non-disruptive capacity expansions.

Sr. NetBackup Consultant — NYC Department of IT & Telecommunications (DoITT)

New York, NY · Sep 2015 – Feb 2016

- **Deployed NetBackup 5230 appliances** across NYC primary data center and New Jersey DR site, engineering clustered **Veritas NetBackup** for RHEL, SUSE, AIX, and Windows protecting municipal Exchange, SharePoint, SQL Server, and Oracle databases.
- **Validated recoverability and audit readiness** through scheduled DR testing on NetBackup with EMC Data Domain, reducing daily backup failure rates from 14% to **under 0.5%** and ensuring 100% municipal audit compliance.

Sr. NetBackup Engineer / Enterprise Technology Consultant — Arrosoft Solutions

New York, NY · Feb 2013 – Jan 2016

Data-protection deployment engineering for **Children's National Medical Center, NYC DoITT, FISA, NJ State Department of Health, Boston EHS, and Yeshiva University / Albert Einstein College of Medicine**.

- **Built multi-site, large-scale data-protection environments** protecting Oracle, SQL Server, Exchange, VMware, and RHEL/SUSE/AIX: Master/Media PDDO/MSDP 5230 appliances, **Auto Image Replication (AIR)**, robotic tape libraries, Storage Lifecycle Policies, SAN storage networks, IPMI, RMAN/SQL/KSH scripting, OpsCenter alerting, and AD/LDAP integration.
- **Implemented EMC VMAX SRDF replication** with active-passive synchronization and Storage Foundation HA / Veritas Cluster Server, validating application and storage failover across primary and DR sites for Children's National Medical Center.
- **Automated hyperscale server deployment** as DevOps Automation Engineer using **Chef, Puppet, PowerShell, and KSH**; built cloud HA failover across AWS EC2/S3, Azure, VMware, and Hyper-V, with centralized syslog, network IDS, and ITIL ticketing.

Earlier Technical Roles

2009 – 2014

- **Hyper-V / Citrix VDI Support Engineer**, Diligex Cloud Services (clients: Viacom, US Space) — managed-service cloud VDI on Hyper-V and Citrix with AWS EC2/S3.
- **Network & Security Engineer**, GetFugu — scalable cloud-rendering infrastructure, intrusion detection, penetration testing.
- **Windows Network Support Analyst**, Star Alliance Capital (40 Wall Street) — trading-floor support for **200+ NYSE high-frequency equity and options traders** (Lightspeed/DAS).
- **Linux/UNIX System Administrator**, Intersessions Hosting — UNIX and Solaris hosting administration.

CERTIFICATIONS & TRAINING

- **Veritas / Symantec**: NetBackup for UNIX, STS Certified (2014) · Symantec Certified Specialist
- **Veeam Certified**: VMSP and VMTSP (2015) | **VMware**: VCP6 (vSphere)
- **NVIDIA Certified (2026)**: AI Factory Deployment Learning Path · Cumulus Linux Essentials · SONiC Essentials · Networking Technical Curriculum
- **Automation & Systems**: IPsoft IPCenter CSI Automation (DevOps) & ICA | Arrosoft Certified Engineer: Backup, Storage & Virtualization

EDUCATION

B.Tech, Computer Information Systems — CUNY, New York City College of Technology, New York, NY (2010)

OPEN SOURCE & PORTFOLIO

- **github.com/senmondip** — publishing an open-source portfolio of enterprise data-protection and infrastructure automation: **38 repositories covering NetBackup, Commvault, IBM Spectrum Protect, Veeam, VMware vSphere, Linux/Windows Server, AWS, storage operations, disaster recovery, and Prometheus/Grafana observability**, released on a scheduled cadence through 2026.
- **Contributor — MH-VTL GUI** (virtual tape library): re-engineered the codebase for release builds.